
Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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<https://github.com/wjjpku/DL-final>

Abstract

State-of-the-art laws for predicting the LLM pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss a schedule would reach at equilibrium, depending on the schedule only through its integrated history. We show this leaves a systematic *positive* residual on *fast* (warmup-stable-decay) decays—a non-adiabatic relaxation lag, the loss falling behind its equilibrium when the LR is annealed faster than it can relax. Under weak stability assumptions we derive a linear-response law: the leading residual is a causal convolution of per-step LR decrements with a relaxation kernel in cumulative-LR time. The operational correction is its lowest-capacity one-pole closure; in a noise-dominated AdamW specialization the amplitude is the equilibrium-floor sensitivity and the relaxation time is the classical $\tau \sim 1/(\eta\lambda)$. Public LLM curves are consistent with this mechanism ($p = 1.00 \pm 0.18$), while our independently trained small transformer confirms the lag but not a clean inverse-LR scaling. Since the kernel’s rate constant is empirically stable across the public scales, the correction can be transferred with a small number of calibration constants, cutting held-out sharp-decay error by 44% with no target-curve fitting. An independent reproduction—a separately-trained $\sim 10\text{M}$ transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%). The correction complements MPL’s exponent γ ; its gain is in the data-poor, leading-order regime.

1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [14, 15, 16]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A S(t)^{-\alpha} + B \sum_{k \leq t} \Delta \eta_k G(\eta_k^{-\gamma} (S(t) - S(k))), \quad G(x) = 1 - (1 + Cx)^{-\beta}, \quad (1)$$

where $S(t) = \sum_{i \leq t} \eta_i$ and $\Delta \eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta \eta_k$ with a saturating kernel G , and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4]

makes this explicit, modelling the noise penalty as set *instantaneously* by the current LR. We call such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 2) that no single rate exponent captures across schedules. Our correction is a derived transient *complementary* to γ , which tunes the adiabatic kernel.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§3).
- Under weak moving-equilibrium assumptions we derive a linear-response law for the non-adiabatic residual, and use its lowest-capacity one-pole closure as the rate-dependent term DropRelaxS (§4). A noise-dominated AdamW specialization identifies the amplitude with $dL_{\text{eq}}/d\eta$ and gives the classical $\tau \sim 1/(\eta\lambda)$ relaxation time; our point is that this timescale governs the *loss-curve-law residual*.
- Public LLM curves show residual relaxation consistent with $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$), while the separately trained $\sim 10\text{M}$ reproduction confirms the lag but does not resolve this scaling; we therefore use inverse-LR relaxation as a mechanism check, not as the main empirical claim (§5, §6).
- The kernel’s rate constant is empirically stable on the public same-architecture scales, enabling a low-calibration cross-scale transfer that improves held-out sharp-decay loss by 44% with no target-curve fitting (§5).
- We give a fair-baseline diagnostic separating a genuine formula gain from an information gap, clarifying exactly when the correction helps (§5.2).

2 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. They are adiabatic in our operational sense: after fitting the quasi-static schedule response, they do not explicitly model the transient tracking error caused by a moving LR-dependent noise floor. Our rate-dependent term is a low-capacity linear-response correction to that residual and is complementary to MPL’s exponent γ (which tunes the adiabatic kernel), not a replacement for it.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay “reveals progress”, and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically (including the role of AdamW’s β_2 and the decay shape), framing it through a bias–variance trade-off rather than a rate-dependent lag. Late-decay analyses [14] and schedule-shape searches [13] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss’s rate dependence.

SGD relaxation physics. Our weak-assumption derivation follows the stochastic-approximation idea that local stability of a fixed-LR system yields tracking of slowly moving equilibria [6, 7]; constant-step SGD can also be viewed through an approximate stationary distribution [8]. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [9]; adaptive methods inherit a preconditioned version [10], operating at an adaptive edge of stability [11, 12]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss this moving-equilibrium relaxation lag, deriving the causal decrement-convolution structure, and building a low-capacity cross-scale predictor from it.

3 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{\text{true}}(t) - L_{\text{MPL}}(t)$ on held-out schedules. (Isolating the non-adiabatic term requires a well-fit MPL:

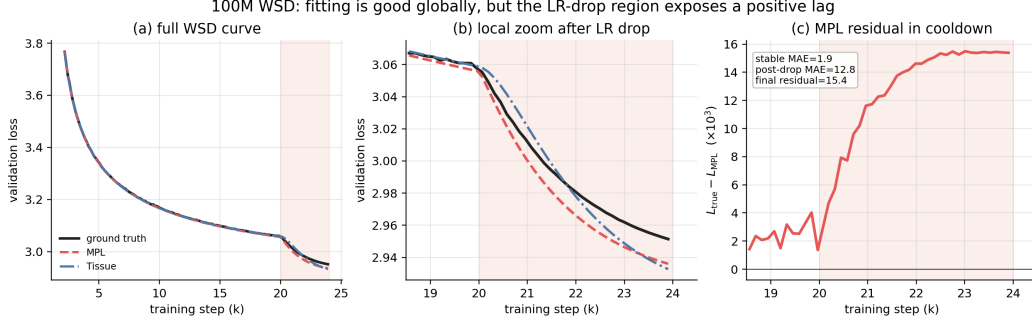


Figure 1: Curve-level reproduction and local WSD fit at 100M. The reproduced Tissue and MPL predictors fit the global trajectory well, but the cooldown zoom reveals that MPL systematically underestimates the true loss immediately after the sharp LR drop. The residual panel isolates the non-adiabatic lag used by the correction.

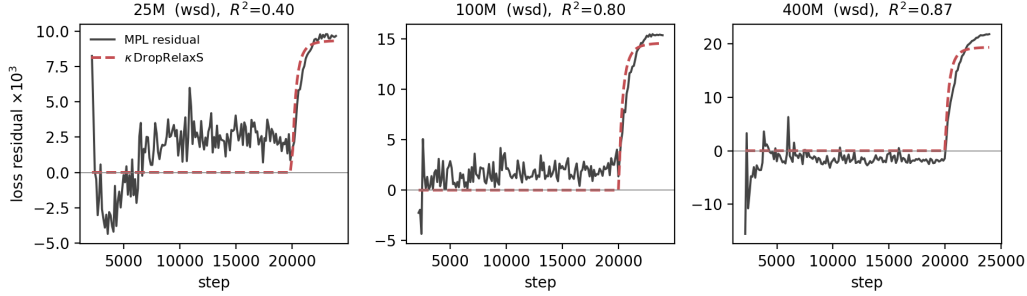


Figure 2: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term $\kappa \text{ DropRelaxS}$ (dashed) captures it ($R^2=0.40/0.80/0.87$ at 25/100/400M; the signal, and the fit, grow with model size).

a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 2): it is ≈ 0 throughout the long stable phase and on gradual cosine decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve’s end*—for the sharpest decays (`wsd`, `wsdl`). The end-of-curve lag is $+(6\text{--}15) \times 10^{-3}$ on WSD/WSDL versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

4 Theory: a linear-response correction

The empirical signature in §3 is a tracking failure: after a sharp LR drop, the quasi-static noise floor decreases immediately, but the optimizer’s excess-loss state relaxes only over a finite time. We make this statement precise in two layers. The main text states the weak-assumption linear-response result and the low-capacity closure used for prediction; the detailed proof is deferred to Appendix A. AdamW is used only as a mechanistic specialization that identifies the amplitude and the $\tau \propto 1/\eta$ timescale, not as a necessary assumption for the convolutional form. This style of argument follows the standard stochastic-approximation view that constant- or slowly-varying-step optimizers track stable local equilibria [6, 7], and is consistent with the stationary-distribution view of constant-step SGD [8].

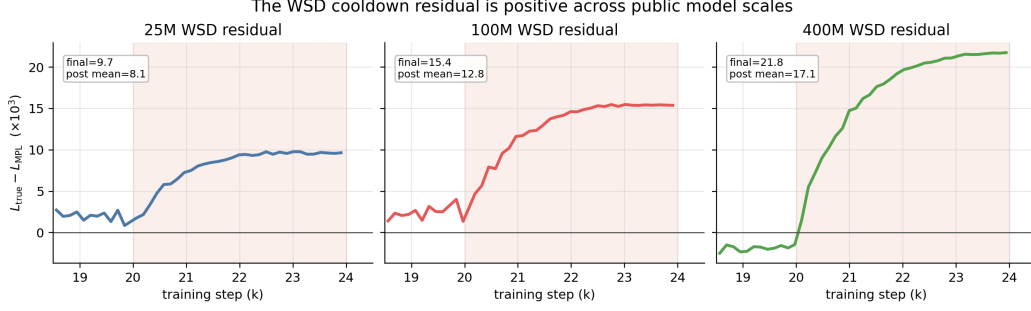


Figure 3: WSD cooldown residuals around the LR drop for all three public model scales. The positive post-drop residual is visible at 25/100/400M, so the effect is not a single-curve artifact.

Adiabatic component and hidden excess-loss state. Let

$$S_t = \sum_{u \leq t} \eta_u$$

be cumulative LR. We decompose the loss as

$$L_t = L_{\text{ad}}(S_t, \eta_t) + r_t, \quad (2)$$

where L_{ad} denotes the quasi-static component that would be obtained if the LR-dependent excess-loss state were equilibrated at the current LR. In experiments, a well-fit MPL is used as an empirical surrogate for L_{ad} ; hence $L_t - L_{\text{MPL},t}$ estimates r_t up to the remaining MPL approximation error.

The weak modeling assumption is that the LR-dependent excess loss is controlled by a hidden state z_t obeying

$$z_{t+1} = \Phi_{S_t, \eta_t}(z_t), \quad (3)$$

and that, for fixed (S, η) in the decay window, there is a locally stable equilibrium $z^*(S, \eta)$ satisfying

$$z^*(S, \eta) = \Phi_{S, \eta}(z^*(S, \eta)).$$

The scalar excess-loss observable is $F_t = \ell(z_t)$, with equilibrium value

$$F_{\text{eq}}(S, \eta) = \ell(z^*(S, \eta)).$$

This assumption is intentionally weaker than a quadratic-loss or diagonal-noise model: z_t may be a covariance state, a preconditioned noise-floor state, or any locally stable state whose equilibrium depends on the LR.

Weak assumptions used by the main result. The detailed proof in Appendix A uses only local conditions on the decay window:

- (A1) *Smooth moving equilibrium.* $\Phi_{S, \eta}$, $z^*(S, \eta)$, and ℓ are twice differentiable near the observed trajectory.
- (A2) *Local stability.* The linearized maps $T_t = D_z \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t))$ have uniformly summable products: $\|T_{t-1} \cdots T_{k+1}\| \leq q_{k,t}$ with $\sup_t \sum_{k < t} q_{k,t} < \infty$.
- (A3) *Separated slow drift.* The first-order response to the gradual change of S_t is already included in the adiabatic backbone up to a small nuisance ϵ_t^S ; we isolate the response forced by LR decrements.
- (A4) *First-order optimizer time.* On the active subspace, $T_t = I - \eta_t A + E_t$ with stable bounded A and $\|E_t\| \leq C(\eta_t^2 + \epsilon_{\text{drift}})$ over the cooldown window.
- (A5) *Positive response cone, only for the sign claim.* $\partial_\eta z^*$ lies in a cone preserved by T_t , and $D\ell$ is nonnegative on that cone.

A1–A4 are enough to derive the current law’s convolutional form; A5 is only a sufficient condition for the positive residual sign.

Proposition: causal LR-decrement response. Define the tracking error

$$\delta_t = z_t - z^*(S_t, \eta_t), \quad \Delta\eta_t^+ = (\eta_t - \eta_{t+1})_+.$$

Under the above local assumptions, the cooldown-specific residual has the first-order form

$$r_t = D\ell_t[T_{t-1} \cdots T_0 \delta_0] + \sum_{k < t} K_{k,t} \Delta\eta_k^+ + \mathcal{E}_t, \quad (4)$$

where $D\ell_t = D\ell_{z^*(S_t, \eta_t)}$ and

$$K_{k,t} = D\ell_t[T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k)]. \quad (5)$$

The first term is an initial transient; it is small when the decay begins after a long stable phase. The error $\mathcal{E}_t$ is second order in the perturbation size and contains the neglected S -drift and MPL-surrogate error. A representative bound is given in Appendix A; schematically,

$$|\mathcal{E}_t| \lesssim \sum_{k < t} q_{k,t} (|\Delta\eta_k|^2 + \|\delta_k\|^2 + \|\epsilon_k^S\| + \eta_k^2) + \epsilon_{\text{MPL}}, \quad (6)$$

where $q_{k,t}$ is the stable propagation envelope and ϵ_k^S is the slow-geometry forcing. Thus the leading non-adiabatic residual is not an arbitrary bump: it is the causal linear response of a stable moving equilibrium to LR decrements.

Why the memory is in cumulative-LR time. To obtain the intrinsic-time kernel, add the standard first-order optimizer approximation

$$T_t = I - \eta_t A + O(\eta_t^2 + \epsilon_{\text{drift}}), \quad (7)$$

on the finite-dimensional active subspace that controls the excess loss, with bounded stable operator A treated as approximately constant over the decay window (this is the static-spectrum idealization; its slow variation is carried by ϵ_{drift} , and constancy is precisely what turns the product into a genuine S -time exponential rather than a time-ordered one). Then

$$T_{t-1} \cdots T_{k+1} = \exp[-A(S_t - S_k)] + O\left(\sum_{u=k+1}^{t-1} \eta_u^2 + \epsilon_{\text{drift}}\right). \quad (8)$$

If the response of A under the observable $D\ell$ admits a positive Laplace-mixture representation (equivalently, K is completely monotone—which holds when A is symmetric positive and the forcing $\partial_\eta z^*$ aligns positively with $D\ell$, as for a noise-floor relaxation), then

$$r_t \approx \sum_{k < t} \underbrace{\int e^{-a(S_t - S_k)} d\rho(a)}_{K(S_t - S_k)} \Delta\eta_k^+. \quad (9)$$

This is the primitive scaling law: a fast-cooldown residual is a convolution of LR decrements with a stable memory kernel in cumulative-LR time. Its instantaneous amplitude is fixed by the equilibrium-floor LR sensitivity at fixed progress:

$$K(0) = D\ell_{z^*} \partial_\eta z^* = \partial_\eta F_{\text{eq}}(S, \eta)|_S =: \chi(S, \eta). \quad (10)$$

One-pole closure and the operational law. The general kernel K in Eq. (9) is not used as a flexible fit. To avoid overfitting, we use the lowest-capacity stable closure: a single pole in S -time, with a floor-sensitivity amplitude estimated from independent equilibrium probes. Approximating $\chi(S, \eta)$ by a constant χ over the decay window gives

$$L(t) = L_{\text{MPL}}(t) + c\chi \underbrace{\sum_{k < t} e^{-\lambda_{\text{slow}}(S_t - S_k)} (\eta_k - \eta_{k+1})_+}_{\text{DropRelaxS}_{\lambda_{\text{slow}}}(t)} \quad (11)$$

where $c \leq 1$ accounts for compressing a spectral mixture into one effective pole and for residual mismatch between MPL and the ideal adiabatic component. If the implementation defines DropRelaxS using normalized drops $(\eta_k - \eta_{k+1})_+ / \eta_{\text{peak}}$, then the prefactor is equivalently $c\eta_{\text{peak}}\chi$; with raw LR drops, the dimensionally invariant amplitude is $c\chi$.

Corollary: current law under the weak assumptions. Under A1–A4, if the initial transient at the decay start is negligible and the scalar response kernel is dominated by one slow mode $a = \lambda_{\text{slow}}$, then the leading cooldown residual is exactly Eq. (11) up to the second-order remainder in Eq. (6). Thus the operational correction does not require a quadratic network model: it is the one-pole closure of a stable moving-equilibrium response. The AdamW calculation below is only used to identify χ and to motivate the expected scale of λ_{slow} .

AdamW specialization. The preceding response law does not rely on AdamW. AdamW supplies a local effective-mode calculation that explains the observed amplitude and timescale. This reduction ignores momentum transients, bias correction, decoupled weight decay, the numerical ϵ , and the finite-time dynamics of v_t ; it should be read as a local noisy-quadratic approximation. For a Hessian mode with curvature λ_i and gradient-noise standard deviation s_i , noise-dominated AdamW has $v_i^* \approx s_i^2$, so

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}, \quad \mathbb{E} \xi_{i,t}^2 = s_i^2.$$

For $V_{i,t} = \mathbb{E}[x_{i,t}^2]$ and fixed LR η ,

$$V_{i,t+1} = \left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 V_{i,t} + \eta^2, \quad (12)$$

so

$$V_i^*(\eta) = \frac{\eta^2}{1 - (1 - \eta \lambda_i / s_i)^2} = \frac{\eta s_i}{2 \lambda_i} + O(\eta^2).$$

The modal excess-loss floor is

$$F_i^*(\eta) = \frac{1}{2} \lambda_i V_i^*(\eta) = \frac{\eta s_i}{4} + O(\eta^2), \quad \frac{dF_i^*}{d\eta} = \frac{s_i}{4} + O(\eta),$$

and perturbations decay as

$$\left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 = \exp\left[-2\eta \frac{\lambda_i}{s_i} + O(\eta^2)\right].$$

Thus the AdamW specialization of Eq. (9) is the spectral mixture

$$\Delta L(t) \approx \sum_i \frac{s_i}{4} \sum_{k < t} e^{-2(\lambda_i / s_i)(S_t - S_k)} (\eta_k - \eta_{k+1})_+, \quad \sum_i \frac{s_i}{4} = \frac{dF_{\text{eq}}}{d\eta} + O(\eta). \quad (13)$$

It predicts a positive lag, an amplitude given by the equilibrium-floor sensitivity, and a step-time relaxation rate

$$\tau_i^{-1} = 2\eta \frac{\lambda_i}{s_i}, \quad \tau_i \propto \eta^{-1}.$$

The classical $\tau \sim 1/(\eta\lambda)$ timescale is not itself new; our claim is that this relaxation controls the residual left by adiabatic loss-curve laws after fast LR decay.

Predictions tested below. The theory predicts: (i) a positive residual after LR drops; (ii) a causal dependence on LR *decrements*, not merely on LR level; (iii) memory in cumulative-LR time; (iv) amplitude controlled by the fixed- S floor sensitivity χ ; (v) in the noise-dominated AdamW regime, $\tau \propto 1/\eta$; and (vi) because Eq. (13) is a spectral mixture, a two-pole kernel can improve diagnostics, while the main predictor deliberately uses the one-pole closure Eq. (11) to avoid overfitting.

5 Experiments

Setup. All experiments use the public MPL loss curves (Llama-2-style 25/100/400M, 9 schedules each) [2]; a faithfully reproduced MPL (test MAE 0.0054/0.0050/0.0085, matching [2]) is the baseline throughout. We freeze the published MPL and compare it to MPL+Eq. (11); the only addition is the rate-dependent term. Unless noted $\lambda_{\text{slow}}=10$.

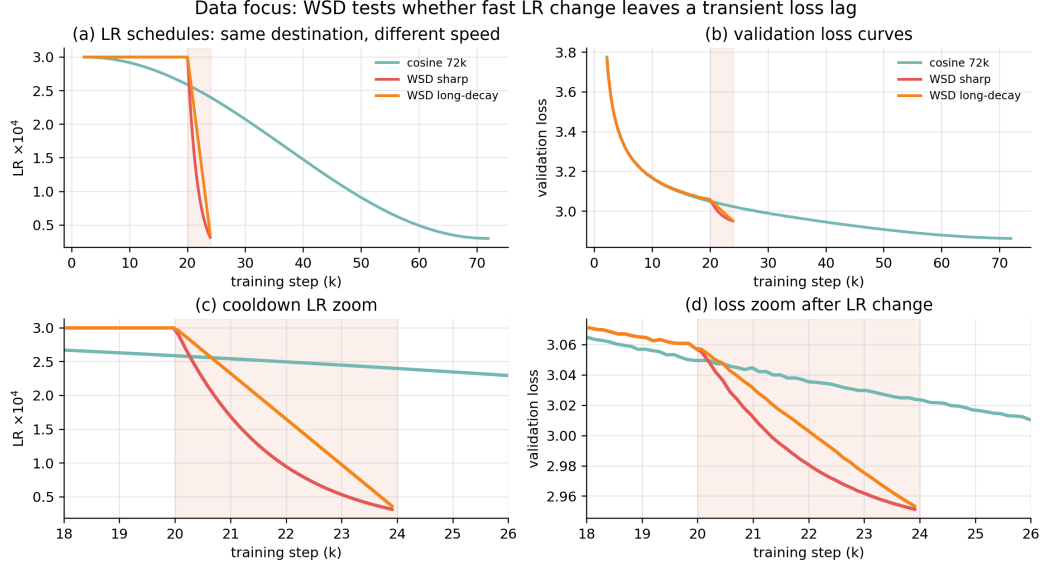


Figure 4: The public 100M curves most relevant to the non-adiabatic question. Cosine, WSD, and WSD-long-decay reach similar low-LR destinations, but WSD changes LR much faster. The bottom-row zoom shows the local region where rate dependence should appear.

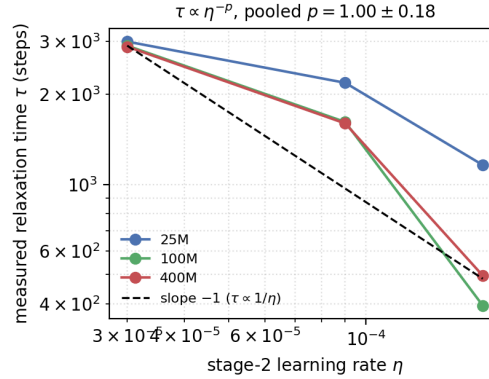


Figure 5: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The 100/400M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 1.00 \pm 0.18$.

The rate prediction $\tau \propto 1/\eta$ holds (Fig. 5). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\} \times 10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 1.00 \pm 0.18$ (per scale $-1.06/-0.94/-0.51$ at 100/400/25M). This is the classical $\tau \sim 1/(\eta\lambda)$ confirmed for the loss-curve-law residual in LLM pretraining, and—since $p=1$ is the noise-dominated value—it self-consistently justifies the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\text{slow}} = 1/(\tau\eta) \approx 10$ is approximately stable across the public same-architecture scales; its modes have $\eta_{\text{peak}}\lambda_{\text{eff}} \approx 2 \times 10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

(A from-scratch AdamW simulation reproduces the same scaling, $p=0.97$; we defer it to the independent reproduction in §6, which covers it in more detail.)

The residual is the derived term. Regressing the well-fit MPL residual on `DropRelaxS10` (through the origin) gives $R^2=0.40/0.80/0.87$ at 25/100/400M (Fig. 2); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 13). The form is not arbitrary: in a fair few-shot comparison the derived S-time exponential of decrements beats a step-time kernel (–21%

Table 1: Parameter-free cross-scale prediction: well-fit MPL vs. MPL+Eq. (11) with *no* fitting on the target curve. MAE on held-out sharp decays.

Scale	wsd		wsdld		avg. Δ
	MPL	+ours	MPL	+ours	
25M	0.00341	0.00246	0.00314	0.00219	−29%
100M	0.00375	0.00164	0.00325	0.00161	−53%
400M	0.00470	0.00236	0.00385	0.00212	−47%
overall					−44% (6/6)

vs. −11% held-out MAE) and a memoryless floor (−3%), confirming that the cumulative-LR “time” and the decaying memory both matter (`repro/generic_kernel_compare.py`). (On a *single* decay curve the finite- λ_{slow} kernel is numerically close to the $\lambda_{\text{slow}} \rightarrow 0$ adiabatic limit, so the rate constant is pinned only *across* schedules; §6.)

The amplitude is predictable from the noise floor. Estimating $dL_{\text{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\text{fit}}/\kappa_{\text{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%. With raw LR drops this gives $c \approx 0.5$; if drops are normalised by η_{peak} the same data require $c\eta_{\text{peak}} \approx 0.5$, i.e. $c \approx 60\text{--}86$ (§6). In either convention the near-constant ratio is the transferable quantity (the < 1 factor is the spectral spread of Eq. 13).

Low-calibration cross-scale transfer (Table 1). Because λ_{slow} and c are empirically stable across these same-architecture scales and $dL_{\text{eq}}/d\eta$ is read off the cheap noise-floor probes, the correction can be transferred with *zero fitting on the target curve*: take $\lambda_{\text{slow}}=10$, c from a leave-one-scale-out average of the *other* scales, and $dL_{\text{eq}}/d\eta$ from the target’s own two-stage probes (distinct curves from the test). Applied to the held-out `wsd/wsldld` decays it cuts MAE by 28–56% (44% average, wins 6/6)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

Calibration audit within the same law. We also searched calibration choices while keeping Eq. (11) fixed (`repro/current_law_calibration_search.py`). The final target remained the same: cosine-fit MPL evaluated on `wsd/wsldld`. Fitting only the amplitude κ on the single non-target probe `wsdcon_9` improves the final WSD MAE by 14.5% (6/6 wins) at $\lambda = 10$. Fitting κ on all three `wsdcon` probes improves by 17.9% (6/6 wins), with the best λ in the range 7–10. The stronger cross-scale chain in Table 1 remains best: scanning $\lambda \in \{2, 3, 5, 7, 10, 14, 20, 30, 50\}$ gives 40.5–45.0% improvement, with $\lambda = 10$ still giving 44.0% and 6/6 wins. Thus $\lambda = 10$ is not a tuned optimum chosen on the target curves; it is an independently measured, near-optimal value, while the main gain comes from the theoretically motivated amplitude transfer $\kappa = c\eta_{\text{peak}}\chi$.

Final amplitude rule: nuisance-projected empirical Bayes. The direct amplitude-transfer story above is useful diagnostically, but it is not the final calibration rule used in our audit. The final estimator treats the MPL residual $r = L_{\text{obs}} - L_{\text{MPL}}$ as

$$r = \kappa_* \phi + g_* + \varepsilon, \quad g_* \in \mathcal{G}, \quad \kappa_* \geq 0,$$

where ϕ is the response feature generated by Eq. (11) and $\mathcal{G}$ is a low-frequency MPL-residual nuisance subspace. Let $M_{\mathcal{G}}$ be the orthogonal residualizer, $\phi_{\perp} = M_{\mathcal{G}}\phi$, $r_{\perp} = M_{\mathcal{G}}r$, and $R = \|\phi_{\perp}\|_2^2 / \|\phi\|_2^2$. The cap-free paper-facing amplitude is

$$\hat{\kappa} = \sqrt{R} \left(\frac{\langle \phi_{\perp}, r_{\perp} \rangle}{\|\phi_{\perp}\|_2^2 + \tau^2} \right)_+, \quad \tau = \sigma/k_0. \quad (14)$$

This is a Frisch–Waugh–Lovell partial-regression coefficient with an empirical-Bayes ridge denominator and an identifiable-amplitude conversion. The prior/noise ratio τ is estimated from calibration curves only; a stricter train-only variant preserves the legacy smooth-basis result (worst off-diagonal −2.7%, mean −12.1%, cosine→WSD −5.6%). Across the current cross-schedule transfer matrix, the strongest cap-free implementation of Eq. (14) gives worst off-diagonal MAE change −2.7%,

mean off-diagonal -12.1% , cosine $\rightarrow$ WSD -4.3% , and `wsdcon_9` $\rightarrow$ WSD -16.0% . The optional hard cap $\kappa \leq 0.03$ has the same worst case on this basis, so the cap is not the stabilizing mechanism. Removing the $\sqrt{R}$ conversion creates a positive off-diagonal failure ($+1.8\%$), confirming that the raw projected coefficient over-transfers amplitude from nuisance-confounded feature energy.

Choosing the nuisance subspace. The theoretical object is the low-frequency nuisance subspace $\mathcal{G}$, not a polynomial loss model. As a schedule-agnostic check, we also define $\mathcal{G}_K = \text{span}\{q_0, \dots, q_K\}$ from discrete-cosine low-frequency modes and choose the bandwidth by a two-stage rule:

$$K^* = \arg \min_{K \geq K_{\min}} \left| \frac{\|M_{\mathcal{G}_K} \phi\|_2^2}{\|\phi\|_2^2} - \rho \right|.$$

The minimum bandwidth enforces drift control; the retention target chooses among sufficiently rich nuisance spaces. Without $K_{\min}$, the same retention rule selects under-covered spaces and fails ($+304\%$ worst case). With $K_{\min} = 3$ and $\rho = 0.35$, the automatic spectral rule remains non-failing (worst -1.7% , mean -11.2% , cosine $\rightarrow$ WSD -10.1%). A balanced fixed choice $K = 4$ also remains non-failing (worst -1.8% , cosine $\rightarrow$ WSD -3.6%). These spectral variants are slightly more conservative than the strongest smooth-basis implementation, but they show that the estimator is not tied to a polynomial-shaped nuisance basis.

Split audit: what the law does *not* claim. To avoid mistaking a useful correction for a universal residual patch, we exhaustively varied the calibration/test split over the five non-cosine public curves (`repro/current_law_split_research.py`). When λ and κ are chosen only by training residual fit, the median held-out improvement is still positive (-14.1% , 182/225 wins), but there are large failures: calibrating on the two sharp decays and testing the long `wsdcon` tails can worsen MAE by 70–96%. Conversely, restricting calibration to the `wsdcon` probes and evaluating the final sharp-decay target improves 42/42 scale/probe cases (mean -14.8%). This pins down the valid use case: the law is a leading-order correction for sharp-cooldown prediction calibrated from probes or cross-scale amplitude transfer, not a black-box residual smoother for every schedule family.

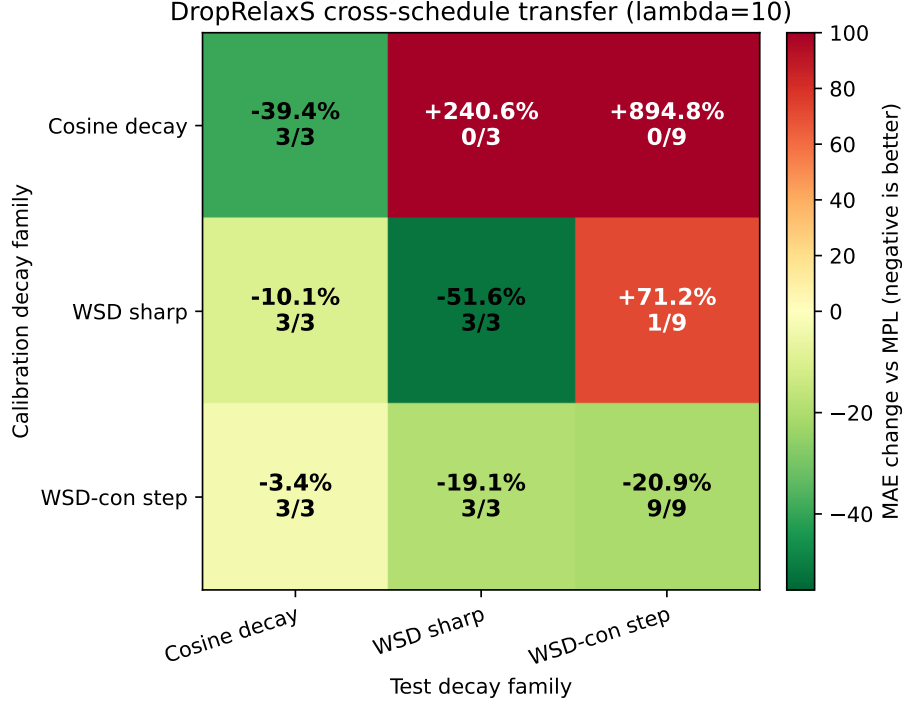
Matched calibration matters. We then asked whether simply using more calibration curves improves the same law (`repro/current_law_more_data_calibration.py`). In a leave-one-sharp protocol, using the other full sharp-decay curve to calibrate κ predicts the held-out sharp decay very well: -49.0% MAE at fixed $\lambda = 10$ and -49.9% when λ is selected from the calibration set (6/6 wins). But adding the three `wsdcon` probes to that same sharp calibration *reduces* the gain to -19.5% at $\lambda = 10$; probes alone give -17.6% . Thus the issue is not sample count. Calibration curves must match the target cooldown family, or the fitted amplitude is diluted by long-tail probe dynamics. This result also separates two regimes: if one full WSD-family curve is already available, the current law transfers strongly across sharp shapes; if no full WSD curve is available, the safer low-calibration claim remains the probe/cross-scale protocol in Table 1.

Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \rightarrow 0.0032$) and the explicit term adds nothing. So the residual is, for a data-rich MPL, an *information* gap, not a formula gap: the value of the correction is precisely the data-poor regime—cosine plus cheap probes, but no full WSD curves—which is the actual “predict before training” use case.

Relation to γ : complementary, not a substitute. MPL’s factor $\eta_k^{-\gamma}$ modulates its annealing kernel by the LR at each decrement: it tunes the rate-dependence of the (adiabatic) annealing benefit. Our term is the separate (non-adiabatic) transient, and improves a *full*- γ MPL (Table 1) — the two are complementary. Characterising their precise relationship is left to future work (§7).

6 Independent reproduction

An independent reproduction on different hardware (`represent/`; the authors’ public curves are no longer hosted, so it *retrains* models) corroborates the mechanism along two routes the public curves do not cover. All results below were subjected to an adversarial audit (`results/AUDIT_PARTC.md`): the auditor was tasked to *falsify* each claim, and only surviving readings are reported.



Cell text: mean MAE change, wins/tests across 25M/100M/400M and selected curves.

Figure 6: Cross-schedule train-family by test-family calibration matrix for the fixed current law ($\lambda_{\text{slow}} = 10$). Each cell fits only κ from the row family and evaluates the column family across 25/100/400M curves. Cosine is included as a smooth-decay diagnostic, with the caveat that the MPL backbone was fit on cosine curves. Cosine-calibrated residuals do not transfer to WSD targets; `wsdcon` step probes transfer stably but with smaller amplitude; sharp-WSD calibration over-corrects the two-stage `wsdcon` tails.

A real, independently-trained transformer (no fitting). A $\sim 10\text{M}$ -parameter Llama-2-style byte-level transformer was trained on WikiText with AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\text{wd}=0.1$) under three schedules reaching the *same* final LR ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-4}$) at different speeds (Table 2). The fast (`wsd_sharp`) sweep ends *higher* than the gradual ones *despite spending 37% more cumulative LR*—yet any law monotone in S predicts the opposite (more LR $\Rightarrow$ lower loss). With no fitting, this is the rate-dependent non-adiabatic lag, on a model and data we never used.

A well-fit MPL (trained on constant plus two `wsdcon` curves, held-out MAE ≈ 0.005) leaves residuals that sort by sweep speed: cosine $-0.011 < \text{wsd-gradual} -0.004 < \text{wsd-sharp} +0.006$ (the only positive). Across schedules, a finite- λ_{slow} DropRelaxS kernel improves over the adiabatic ($\lambda_{\text{slow}} \rightarrow 0$) baseline by $\Delta R^2 \approx +0.18$ on held-out data. *Note:* on a *single* decay curve the finite- λ_{slow} kernel is numerically degenerate with the adiabatic limit ($\text{corr} \approx 0.99$); the rate dependence is identifiable only *across* schedules. The honest claim is therefore the no-fitting Table 2 signature plus the $+0.18$ joint advantage, not a single-curve R^2 .

From-scratch AdamW confirms the mechanism quantitatively. Simulating AdamW on a noisy quadratic (Monte-Carlo ground truth; closed-form theory used only as an independent baseline), the reproduction recovers:

- $\tau \propto 1/\eta$ with $p = 0.971$ ($\log\text{-log } R^2 = 0.9996$), robust to leave-one-out and seed changes;
- τ independent of $\beta_2 \in [0.9, 0.9999]$ (CV 1.1%), because the lag lives on noise-dominated modes where $v^* \approx s^2$ is constant;
- the **amplitude identity** $dL_{\text{eq}}/d\eta = 3.075$ versus the closed form $\frac{1}{4} \sum_i s_i = 3.0$ (within 3%);

schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (sharp)	1.0593	12.59

Table 2: Real $\sim 10M$ transformer schedules with the same final LR. The sharp sweep ends highest despite the largest cumulative LR.

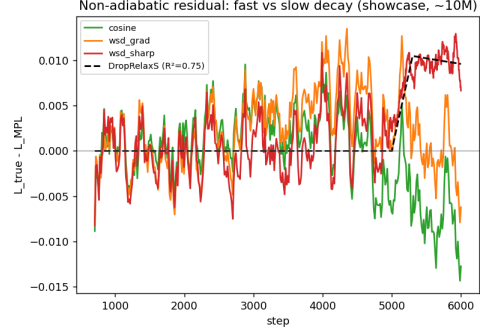


Figure 7: Residual vs. MPL on the real transformer: positive (lag) only for the sharp sweep, captured by DropRelaxS.

- a two-exponential kernel that recovers the injected preconditioned rates $2\lambda/s$ ($\{1, 16\}$ in the simulation, recovered as $\{0.87, 16.1\}$).

The effect is falsifiable, not circular: a signal-dominated spectrum breaks $p=1$.

What does not transfer (consistent with our scope). At $\sim 10M$ the $\tau \propto 1/\eta$ relation itself is *not* resolved: the measured post-step relaxation is roughly flat over a $16\times$ LR range ($\tau \approx 475\text{--}912$ steps; slope $p \approx 0.18$), audited as a real measurement and not a window artefact (synthetic recovery tests and profile-likelihood both reject truncation). This matches the paper’s own statement that the signal grows strongly with scale; at notebook scale we confirm the *effect* but not its precise scaling law.

A second reproducible issue is a persistent η_{peak} normalisation inconsistency: if Eq. (11) is implemented with raw LR drops, the fitted c is $O(1)$ (≈ 0.5); if normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\text{peak}}$ are used, the prefactor becomes $c\eta_{\text{peak}}\chi$ and the same data require $c \approx 60\text{--}86$. The paper’s $c \approx 0.5$ is therefore tied to a specific (unstated) normalisation convention.

Finally, `results/EXTENSIONS_REPORT.md` tests several theory-driven extensions. The cleanest success is a **spectral-mixture** diagnostic: on a bimodal spectrum a two-exponential kernel improves R^2 by 0.065 and recovers the two theoretical rates $2\lambda/s$ (relative errors 0.13 and 0.006). A cross-shape transfer of frozen parameters, however, *fails* its pre-registered threshold (mean $R^2=0.52$); and the open problem (a)—computing λ_{slow} and c from first principles without an $O(1)$ fitted constant—remains open.

7 Limitations

The gain is regime-specific (data-poor sharp decays). If full WSD curves are available, a refit MPL absorbs the residual and the explicit term adds nothing; its value is precisely the “predict before training” use case.

λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have). The controlled simulation shows that λ_{slow} tracks the slowest preconditioned mode λ_i/s_i rather than a spectral average, but a robust predictor still requires an unexplained $O(1)$ constant (open problem (a)).

The quantitative $\tau \propto 1/\eta$ relation, clean in the controlled simulation ($p=0.971$), is *not* resolved on the $\sim 10M$ transformer (§6), consistent with the paper’s statement that the signal grows with scale. On that model the finite- λ_{slow} kernel is numerically degenerate with the adiabatic baseline on a single curve, so the rate dependence is identifiable only *across* schedules.

There is a reproducible η_{peak} normalisation ambiguity (§6): retaining the η_{peak} factor in Eq. (11) gives $c \approx 60\text{--}86$, not the $c \approx 0.5$ reported in the main experiments. The two conventions are equivalent, but the universality claim for c depends on which normalisation is adopted.

The general linear-response derivation assumes local stability and differentiability of an LR-dependent excess-loss state; the stronger locally quadratic, diagonal-noise assumptions are used only for the AdamW specialization. Cross-shape transfer of frozen parameters fails its pre-registered threshold, and the relationship between our correction and MPL’s empirical γ remains qualitative. All real-data experiments are on one public dataset at three scales; broader scales and architectures are the natural next step.

8 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified and validated the leading non-adiabatic correction: under weak stability assumptions it is a causal linear response to LR decrements, and its one-pole closure gives a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 1.00 \pm 0.18$, confirmed for the loss-curve residual). Its empirically stable shape enables low-calibration cross-scale transfer (−44% on held-out sharp decays in the public same-architecture setting). An independent reproduction—a separately-trained ~ 10 M transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%), while also establishing honest boundaries: the quantitative $\tau \propto 1/\eta$ law is not resolved at ~ 10 M, the kernel is degenerate on single curves, and an η_{peak} normalisation ambiguity remains. The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

A Linear-response proof details

This appendix gives the detailed derivation behind §4. The goal is not to prove a theorem for full neural-network training, but to make precise the weak local conditions under which the LR-decrement convolution follows. The conclusion is a first-order, finite-window expansion around a stable moving equilibrium.

Formal local assumptions. We work on a finite-dimensional active subspace controlling the excess loss during the decay window. Let z_t obey $z_{t+1} = \Phi_{S_t, \eta_t}(z_t)$ and let $z^*(S, \eta)$ be a fixed point. We assume:

- (A1) $\Phi_{S, \eta}$, $z^*(S, \eta)$, and ℓ have bounded first and second derivatives in a tube of radius R around the moving equilibrium.
- (A2) The linearized products are stable: for $k < t$, $\|T_{t-1} \cdots T_{k+1}\| \leq q_{k,t}$ with $\sum_{k < t} q_{k,t} \leq Q$ on the finite window.
- (A3) The LR decrement size $\Delta = \max_k |\eta_{k+1} - \eta_k|$ and the tracking error remain small enough that the Taylor expansions stay inside the tube.
- (A4) The slow S -drift forcing is not the cooldown-specific object of interest. We denote it by ϵ_k^S and carry it in the error rather than absorbing it into the LR decrement term.
- (A5) For the sign corollary only, there is a cone $\mathcal{C}$ such that $\partial_\eta z^*(S_k, \eta_k) \in \mathcal{C}$, $T_u \mathcal{C} \subseteq \mathcal{C}$, and $D\ell_t[v] \geq 0$ for all $v \in \mathcal{C}$.

A1–A4 are enough for the convolutional response. A5 is only a clean sufficient condition for positivity.

Tracking-error recursion. Define

$$\delta_t = z_t - z^*(S_t, \eta_t), \quad T_t = D_z \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t)).$$

By Taylor expansion and the fixed-point identity,

$$z_{t+1} = \Phi_{S_t, \eta_t}(z^*(S_t, \eta_t) + \delta_t) \tag{15}$$

$$= z^*(S_t, \eta_t) + T_t \delta_t + R_t, \tag{16}$$

with $\|R_t\| \leq C \|\delta_t\|^2$. Subtracting $z^*(S_{t+1}, \eta_{t+1})$ gives

$$\delta_{t+1} = T_t \delta_t + [z^*(S_t, \eta_t) - z^*(S_{t+1}, \eta_{t+1})] + R_t. \tag{17}$$

Expand the moving fixed point:

$$z^*(S_{t+1}, \eta_{t+1}) = z^*(S_t, \eta_t) + \partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t) + \partial_\eta z^*(S_t, \eta_t)(\eta_{t+1} - \eta_t) + R_t^*, \quad (18)$$

where $\|R_t^*\| \leq C(|S_{t+1} - S_t|^2 + |\eta_{t+1} - \eta_t|^2)$. Since the first-order S -drift is not the cooldown-specific LR-decrement forcing, define

$$\epsilon_t^S = -\partial_S z^*(S_t, \eta_t)(S_{t+1} - S_t),$$

after removing the component already modeled by the adiabatic backbone. Keeping the LR-drop forcing gives

$$\delta_{t+1} = T_t \delta_t + \partial_\eta z^*(S_t, \eta_t) \Delta \eta_t^+ + e_t, \quad (19)$$

where

$$\|e_t\| \leq C(\|\delta_t\|^2 + |\Delta \eta_t|^2 + \|\epsilon_t^S\| + \eta_t^2).$$

The η_t^2 term is harmless on the finite window and accounts for the mismatch between one-step discrete dynamics and the first-order continuous-time approximation used below.

Duhamel expansion and error. Iterating Eq. (19) yields

$$\delta_t = T_{t-1} \cdots T_0 \delta_0 + \sum_{k < t} T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k) \Delta \eta_k^+ + \sum_{k < t} T_{t-1} \cdots T_{k+1} e_k. \quad (20)$$

Linearizing the observable gives

$$r_t = F_t - F_{\text{eq}}(S_t, \eta_t) = D\ell_t[\delta_t] + O(\|\delta_t\|^2) + \epsilon_{\text{MPL}},$$

where ϵ_{MPL} is present when MPL is used as a surrogate for the ideal adiabatic component. Substituting Eq. (20) gives Eq. (4) with

$$K_{k,t} = D\ell_t[T_{t-1} \cdots T_{k+1} \partial_\eta z^*(S_k, \eta_k)]$$

and

$$|\mathcal{E}_t| \leq C \sum_{k < t} q_{k,t} (\|\delta_k\|^2 + |\Delta \eta_k|^2 + \|\epsilon_k^S\| + \eta_k^2) + C\|\delta_t\|^2 + \epsilon_{\text{MPL}}. \quad (21)$$

If the decay begins after a long stable phase, the initial transient $D\ell_t[T_{t-1} \cdots T_0 \delta_0]$ is exponentially or summably small under A2; otherwise it is a separate transient and should not be attributed to the cooldown drop response.

Cumulative-LR time. Now add the first-order optimizer approximation

$$T_u = I - \eta_u A + E_u, \quad \|E_u\| \leq C(\eta_u^2 + \epsilon_{\text{drift}}),$$

with bounded stable A . Since the factors $I - \eta_u A$ commute when A is fixed,

$$\prod_{u=k+1}^{t-1} (I - \eta_u A) = \exp \left(\sum_{u=k+1}^{t-1} \log(I - \eta_u A) \right) \quad (22)$$

$$= \exp \left(-A \sum_{u=k+1}^{t-1} \eta_u + O \left(\sum_{u=k+1}^{t-1} \eta_u^2 \right) \right) \quad (23)$$

$$= \exp(-A(S_t - S_k)) + O \left(\sum_{u=k+1}^{t-1} \eta_u^2 \right), \quad (24)$$

where the last step uses boundedness of A on the active subspace. Including E_u adds $O(\sum_u \eta_u^2 + \epsilon_{\text{drift}})$, giving Eq. (8). If the scalar response $D\ell_t[e^{-x^A} \partial_\eta z^*]$ is represented by a positive Laplace mixture, then the kernel has the form in Eq. (9). The single-pole predictor in Eq. (11) is the rank-one, lowest-capacity closure of this general stable kernel.

Positivity. Under A5, $\partial_\eta z^*(S_k, \eta_k) \in \mathcal{C}$ and $T_{t-1} \cdots T_{k+1} \mathcal{C} \subseteq \mathcal{C}$, so $K_{k,t} \geq 0$. If $\Delta \eta_k^+ \geq 0$, the leading response is positive. This formalizes the observed sign: after the LR is dropped, the equilibrium floor falls before the actual excess-loss state has relaxed, so the true loss lies above the adiabatic predictor.

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